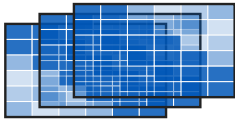
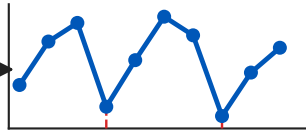


## Step 3 - matrix to vector



sum  $r^2$   
per SNP



diagonal-sum vector